NIHVA

INSTALLATION & MAINTENANCE MANUAL

FOR

DOCK LEVELER

CLIENT: - MUKUND CONSTRUCTION.
PROJECT: - BWF TEC INDIA PVT. LTD.

MFG.YEAR: - APRIL 2017

DESIGNED, DEVELOPED & MARKETED

BY

NIHVA Technologies Pvt. Ltd.

G-BLOCK, 1ST FLOOR, THERMAX CHOWK
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DOCK LEVELER	MANUAL

A DISPATCH & UNLOADING

1. Loading, for dispatch, will be our responsibility. On receipt of the Dock-leveler at site, please check and ensure that there is no physical damage to the equipment.

2. While unloading the equipment from the transporting vehicle, please ensure to use slings with rated capacity higher than the gross weight of the Stationary lift in packed condition.

3. Slings MUST be hooked ONLY at specified positions. These positions are indicated and marked with a figure of 'HOOK'. Improper sling position or improper slinging could cause damage to the equipment.

B. ORDERING OF SPARES

1. When ordering for spare, please specify the following in your Purchase Order.

   - Equipment Name
   - Model Number
   - Equipment Serial No.
   - Part name / Description
   - Part Number
   - Quantity required
   - Delivery period / time
   - Mode of dispatch

   This would avoid ambiguity and help us serve you better.

DATASHEET FOR DOCK LEVELER

CLIENT:

1.0 CONSTRUCTION DETAILS

1.1	Required Capacity - each	: 6 ton
1.2	Model No.	: NDL848
1.3	Leveler operation mode	: AC power pack hydraulically operated
1.4	Lip operation mode	: hydraulic
1.5	Operation control	: by push buttons and control panel
1.6	Loading platform height from road Level	: 1250 mm (Will be confirmed During Detailing)
1.7	Travel above & below dock level	: (+) 300 / (-) 250 mm
1.8	Platform Size (L x W)	: 4000 x 2100 mm
1.9	Lip size (L x W)	: 4000 x 400 mm
1.10	Working length	: 4400 mm
1.11	Overall length	: 4000 mm
1.12	Maximum slope can be achieved	: 8 to 10%
1.13	Clear Pit dimensions (W x Lx D)	: 2150x 4100 x 485 mm
1.14	Hydraulic power pack	
1.14.1	Make	"ASK"
1.14.2	Driver type	: Direct by electric Motor
1.14.3	Absorbed power	: 0.75 kW
1.14.4	Motor Ratings	: 0.75/1 kW /HP
1.14.5	Degree of protection for motor	: IP 55
1.14.6	Class of Insulation for motor	: Class F with temp. Rise limited to Class B
1.14.7	Motor RPM	: 1410
1.14.8	Pump design flow	: 2.5 LPM
1.14.9	Pump pressure	: 120 bar
1.14.10	Safety valve set pressure	: 200 bar or Kg/cm²
1.14.11	in built safety valve	: Yes
1.14.12	Pump type	: Gear
1.14.13	Valves required	: By vendor
1.14.14	Volume of oil receiver	: 07 Lit
1.14.15	Dimensions of power pack Unit	: 350 H x 600 L x 325 W mm
1.14.16	First fill of oil / Qty	: By Vendor
1.14.17	Hydraulic tubing	: By Vendor
1.14.18	Instrumentation	: By vendor
1.14.19	Control panel	: By vendor
1.15	Overall dimensions (l x w x h)	: 2100 x 4000 X 480 mm
1.16	Empty weight	: 1500 Kg

DOCK LEVELLER

INTRODUCTION: -

The hydraulic Dock Leveler is the result of many years' experience in materials handling engineering. “NIHVA” offers the best possible factor of safety with simple, easy to use controls. The high quality of manufacture gives a robust construction and many years of useful life. Each lip and platform movement can be individually controlled (or corrected) by push button.

The dock leveler can be set at any position. During loading, the Dock Leveler is in floating mode i.e. it follows the up/down movements of the vehicle deck. Even with lopsided vehicles, the Dock leveler lip will remain in contact, and toe-traps are avoided by the tilt built into the Dock Leveler.

The whole platform is designed to flex laterally adequately to adapt to sideways tilted vehicles. The leveler always seals perfectly to the lorry bed.

PRINCIPLE: -

Dock Leveler is an adjustable platform used between the transport vehicle and dispatch area of the warehouse of factory for efficient and safe material transfer applications. It bridges the gap between the dispatch area and the transport vehicle and compensates for variations in the truck bed height.

CONSTRUCTION: -

It basically consists of heavy duty fabricated Top Chequered platform and bottom angle channel construction frame. Both are joined by stainless steel machined pins. Lip is hinged to the top platform by means of stainless steel pins.

Top Platform is reinforced with I Beam welded below the platform to strengthen the platform. The cylinder is mounted at the best advantage point against the top platform and bottom frame. Compact power pack includes a single HP motor with overload protection. Hydraulic Pump is Dowty make. Hydraulic operations are by using solenoid control valve.

FEATURES: -

1. Convenient location of all controls.
2. Independent deck and lip.
3. Controls for quicker cycling time.
4. Smooth dependable hydraulics.
5. Quite operation at the touch of the button.
6. Years of trouble-free service.
7. Lower lift time cost and fewer maintenance requirements.
8. Recommended for all applications.
9. Maintenance Strut.
10. Hazard Stripes.
11. Toe guards.

PRE – OPERATING INSTRUCTIONS: -

1. Ensure Hydraulic Oil in the power pack.
2. Ensure Motor connection and proper earthing.
3. Ensure joint leakage.
4. Ensure proper location of platform and lip.

ROUTINE MAINTENANCE AND LUBRICATION

These recommendations are applicable to the Dock Leveler considering average daily usage of 15 to 20 times. If it is used more frequently, the same time intervals for the maintenance should be adjusted accordingly.

DAILY: -

1. Check Hydraulic pipe lines for leak proof working.
2. Check electrical connections for their proper working.

MONTHLY: -

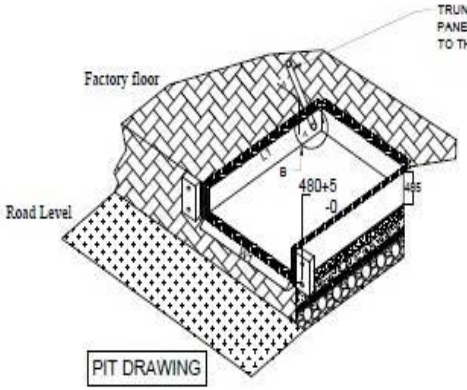
1. Grease the hinge pins at bottom frame and top plate joint and extension lip plate hinge joint.
2. Grease the main Jack Cylinder and Lip Jack Cylinder, Front and Rear Trunion bearings.
3. Change Hydraulic Oil.

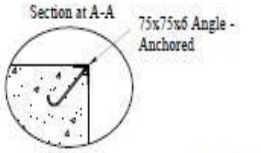
ASSEMBLIES
&
SPARE PARTS LIST

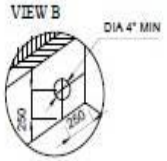
DOCK LEVELER	MANUAL

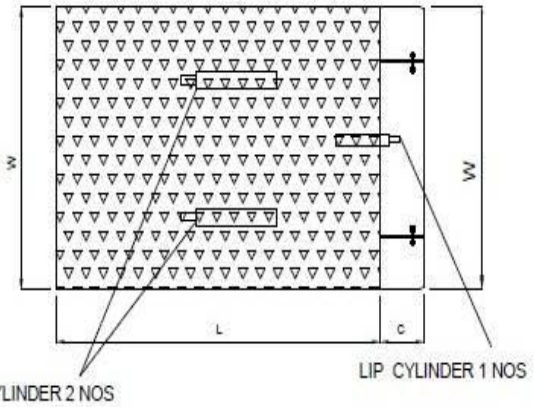
DOCK LEVELER SIZE	WIDTH (W)	LENGTH (L)	HEIGHT (H)	LIP (C) LENGTH	BUFFER SIZE (OPTIONAL)
	2100	4000	480	400	200X300X100

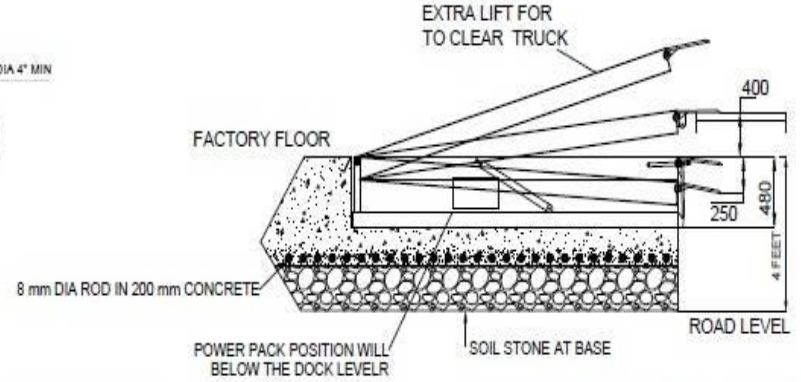
PIT DIMENSION	WIDTH (W 1)	LENGTH (L 1)	DEPTH (D)	ALL DIMENSIONS ARE IN mm
	2150	4100	485	









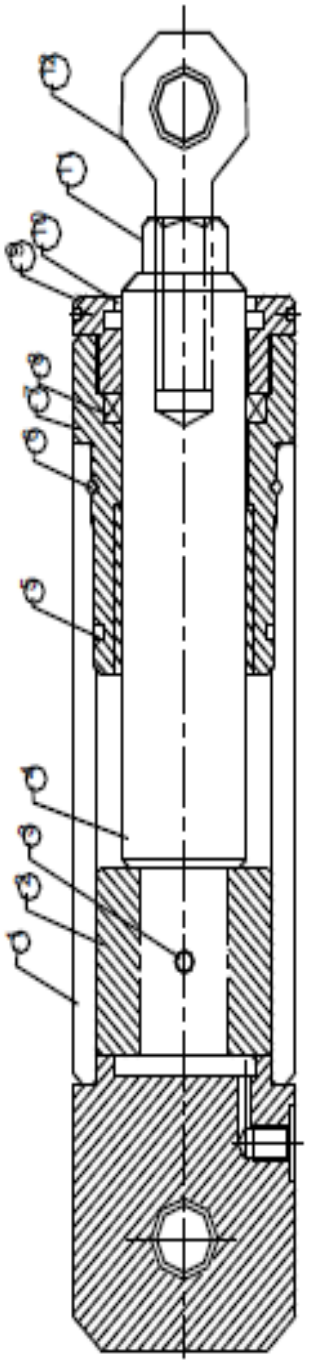


NOTE: -
1. PROPER DRAINAGE SHOULD BE DONE
2. TRENCH AS SHOWN SHOULD BE MADE BY DECIDING THE POWER PACK POSITION (SUPPLY CABLE LENGTH STD. 7000 MM)
3. LEVEL OF THE PIT SHOULD BE PLAIN AND SHOULD TAKE AT LEAST 12000 KG FLOOR LOADING
4. MOUNTING ARRANGEMENT FOR BUFFERS SHOULD BE MADE AS PER BUFFER DRAWING (IF ORDERED)
5. FABRICATION TOLERANCE FOR PLATFROM SIZE WILL BE ± 15 MM

DOCK LEVELER- 06 TON CAPACITY -

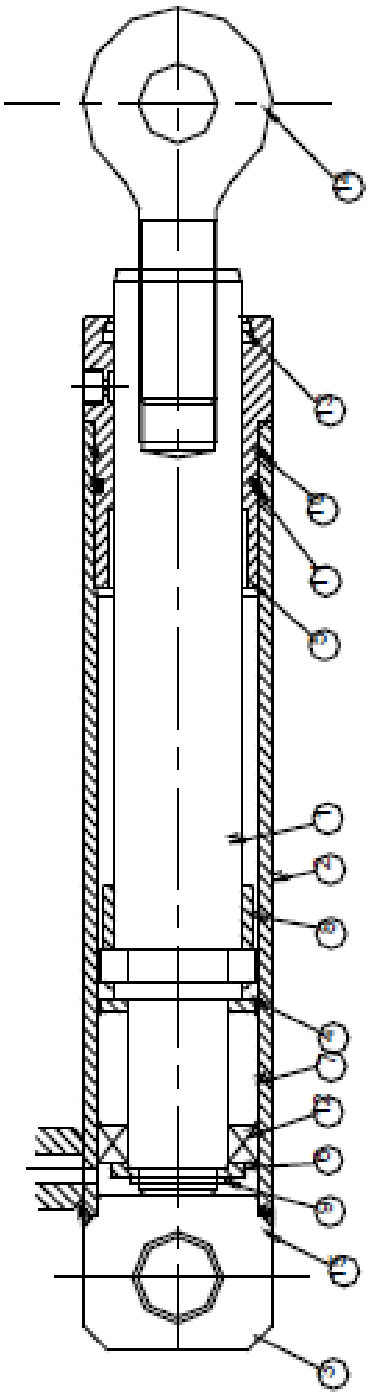
CLIENT APPORVAL-			NIHVA TECHNOLOGIES PVT. LTD.
	DATE	SIGN.	
DRAWN :- OMKAR	03/03/2017		CLIENT - MUKUND CONSTRUCTION
CHECKED :- SAGAR P.	03/03/2017		PROJECT -
ALL DIMENSION ARE IN mm			SITE -
If any queries contact : projects@nihva.com			DWG. NO. - NIHVA/16-17/163

NIHVA TECHNOLOGY PVT. LTD. PUNE
Tel : 020 66304033
Email :- info@nihva.com
Website :- www.nihva.com



SR.NO.	PARTS NO.	DESCRIPTION	QTY
12	REB00006	ROD END BEARING	1
11		HEX NUT M-16X2P	1
10	WPS00003	SEAL WIPER	1
9	70000390	WIPER SEAL HOUSING	1
8	RDS00003	SEAL ROD	1
7	70000389	SEAL HOUSING	1
6	LKW13235	WIRE LOCKING	1
5	ORN00002	O RING	1
4	70000388	SHAFT(PLATED)	1
3	DSP00001	SPRING PIN	1
2	70000387	PISTON	1
1	70000386	JACK CYLINDER	1

DETAILS:-8
LIP JACK ASSLY
PART NO-70000383



SR.NO.	ITEM NO.	PARTS NO.	DESCRIPTION	SIZE	QTY
15	15	70000071	SINTERED BUSH	1"X1.1/8"X1"	2
14	14	70000070	ROD END BEARING	M24X2X25PIN	1
13	13	WPS00015	WIPER SEAL	Ø40xØ48x7	1
12	12	PTS00011	PISTON U SEAL	Ø50xØ40x9	1
11	11	ORN00040	O RING	Ø48X2.5	1
10	10	LKW13225	LOCKING WIRE	13 GAUGE	1
9	9	ECP00009	EXTERNAL CIRCLIP	EXT Ø25	1
8	8	70000050	SPACER Ø40	Ø40XØ49XSP	1
7	7	70000154	PISTON Ø40	Ø50.8XØ31.75X50	1
6	6	70000049	WAHSER RETAINER SEAL	Ø48XØ25.2X5	1
5	5	70000048	WIPER SEAL HOUSING	Ø63X85 L	1
4	4	70000047	WASHER	Ø49XØ32X4	1
3	3	70000159	BOTTOM END	Ø63X49	1
2	2	70000158	ASSLY WELDED CYLINDER	2" HONED	1
1	1	70000157	SHAFT JACK	Ø40 x 580	1

DETAILS:-3
MAIN JACK ASSLY
PART NO-70000156

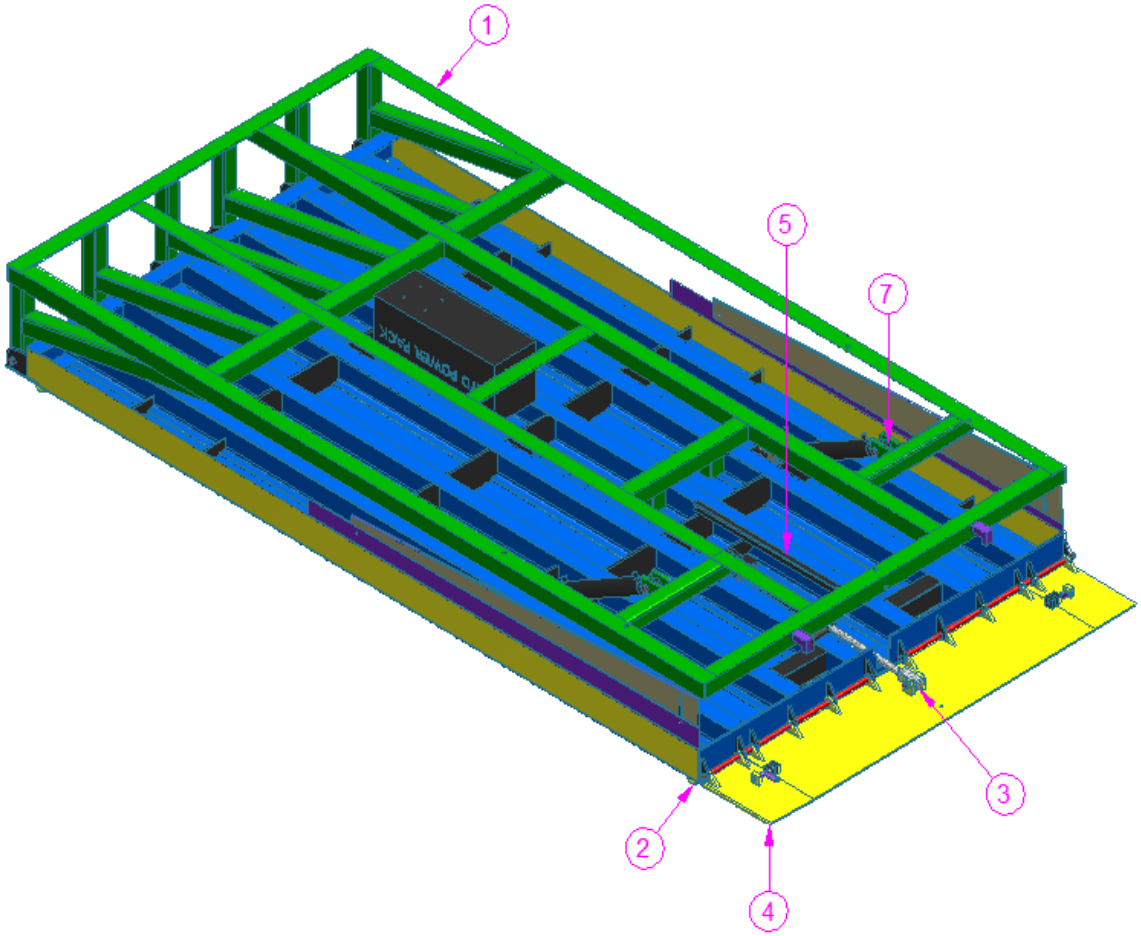
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Mechanical Critical Bought-out Parts

S NO	ITEM DESCRIPTION	QTY.	UNITS
1	HYD-Ø63-Ø-400	2	
2	HYD-Ø50-Ø36-350	1	

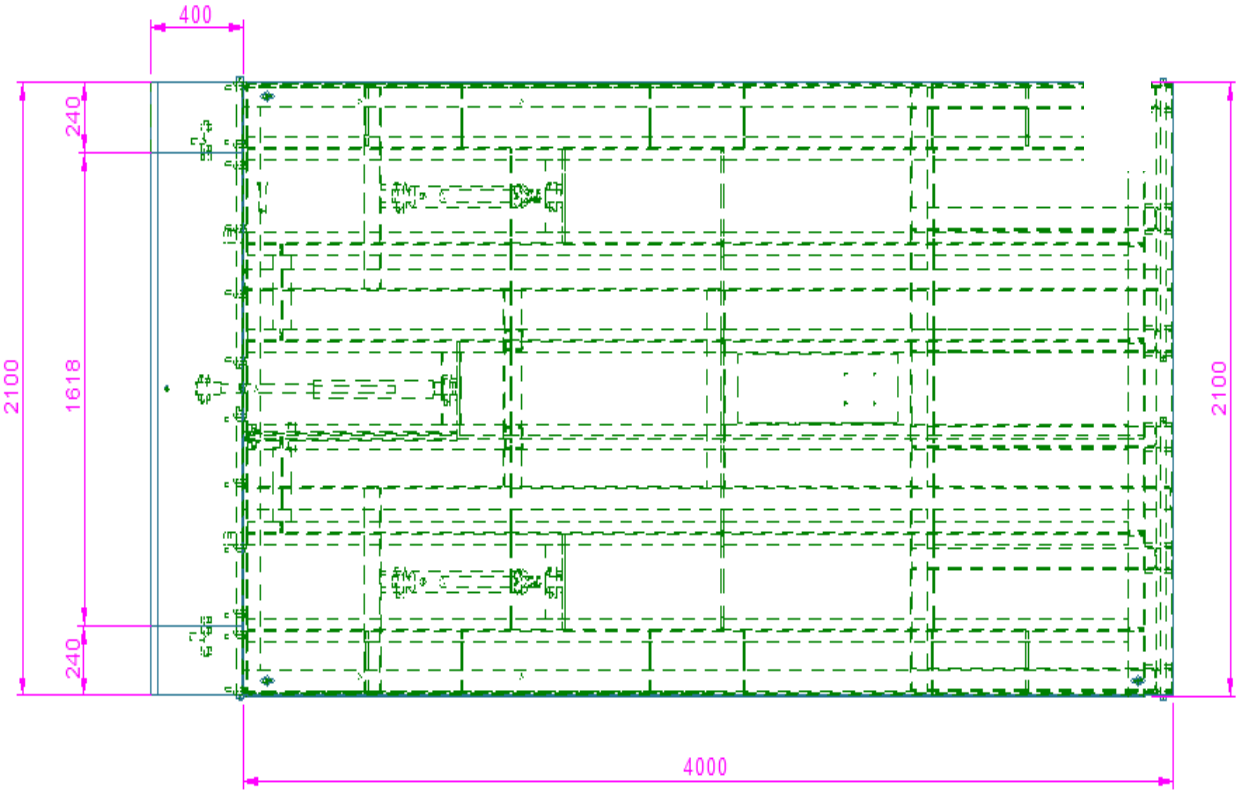
Mechanical Wear Parts

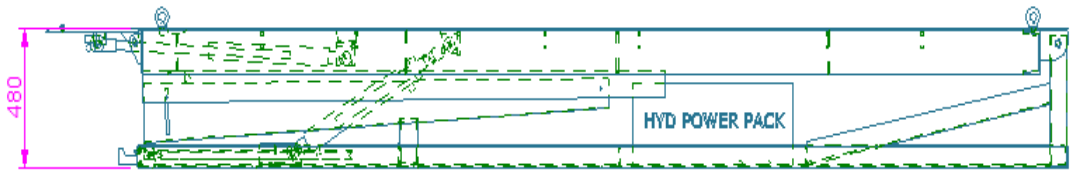
Sr. No.	Units / Part code	Units / Part Description	Qty.
1	170006010103	PIN-1	3
2	170006010201	PIN	2
3	170006010301	PIN-1	2



SR NO	ITEM CODE	DESCRIPTION	QTY.
2	17000601U001	BOTTOM FRAME UNIT	1
3	17000601U002	TOP FRAME UNIT	1
4	17000601U003	HYD CYL UNIT-2	1
5	17000601U004	LIP UNIT	1
6	17000601U005	MANUAL STOPER UNIT	1

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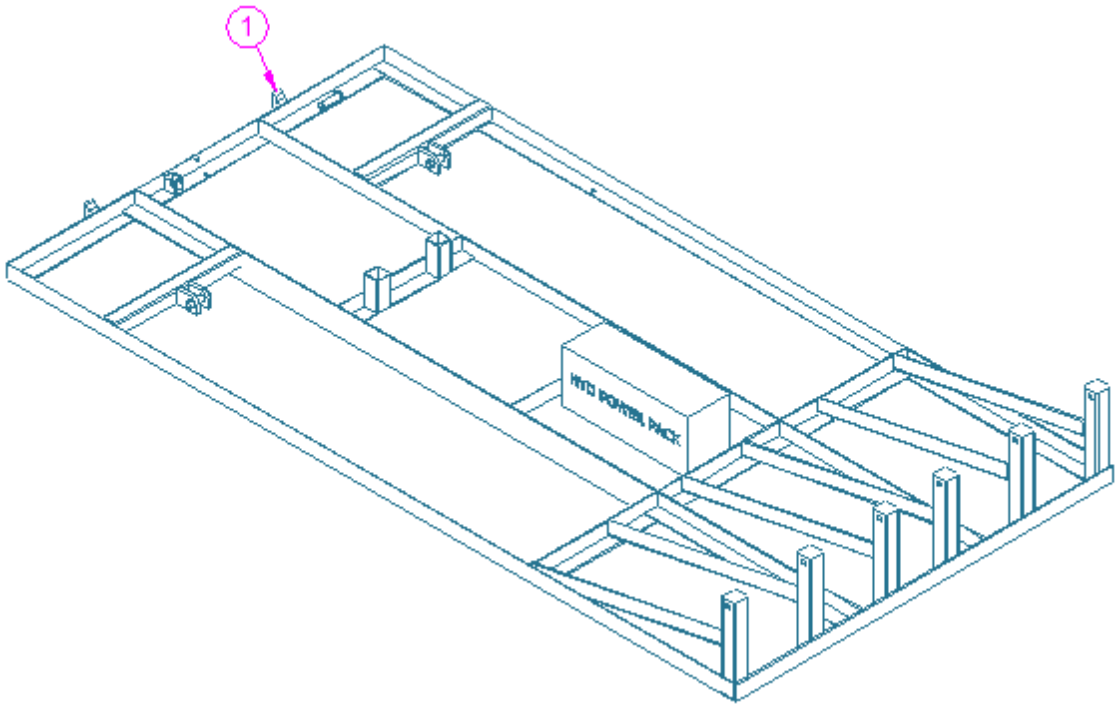




Over all Dimensions views.

DOCK LEVELER	MANUAL

Bottom Frame Unit



BOTTOM FRAME UNIT

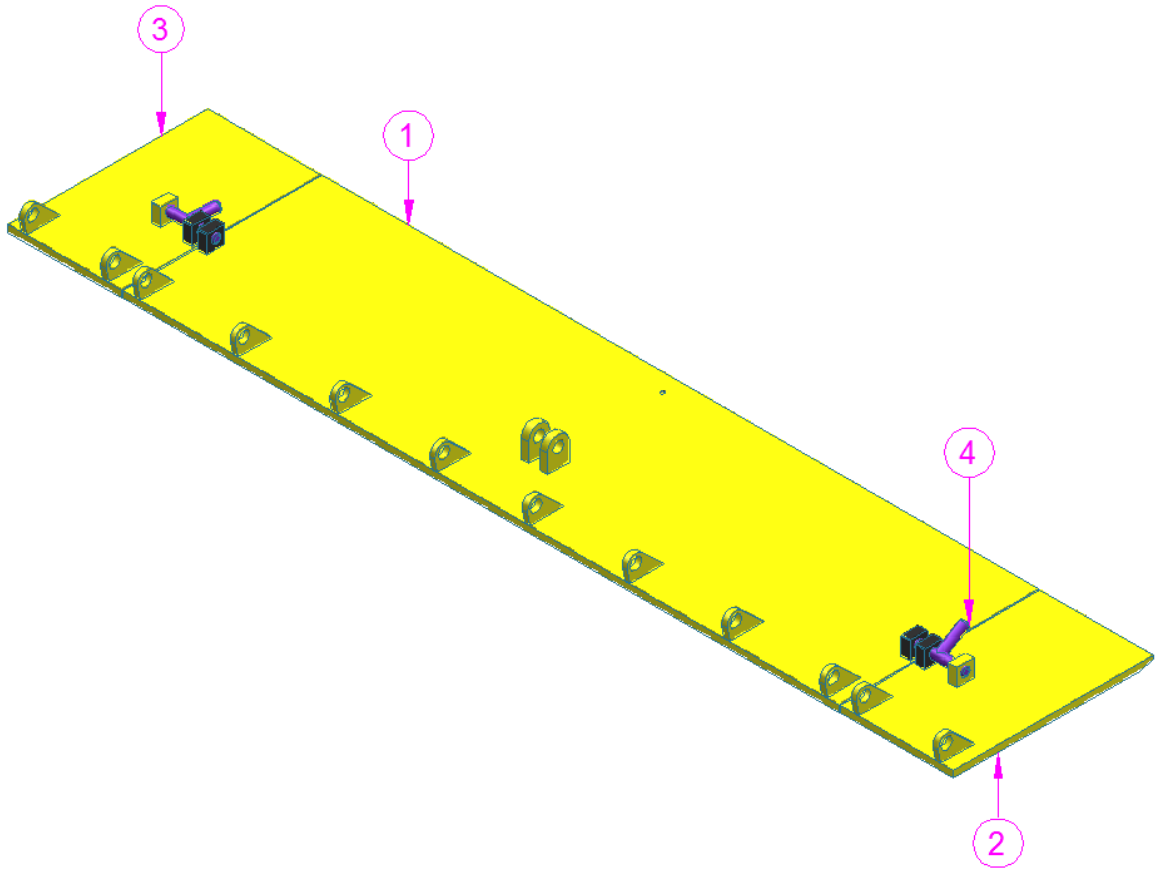
SR NO.	ITEM CODE	DESCRIPTION	QTY
1	170006010101	BOTTOM FRAME	1

Manual Stopper Unit



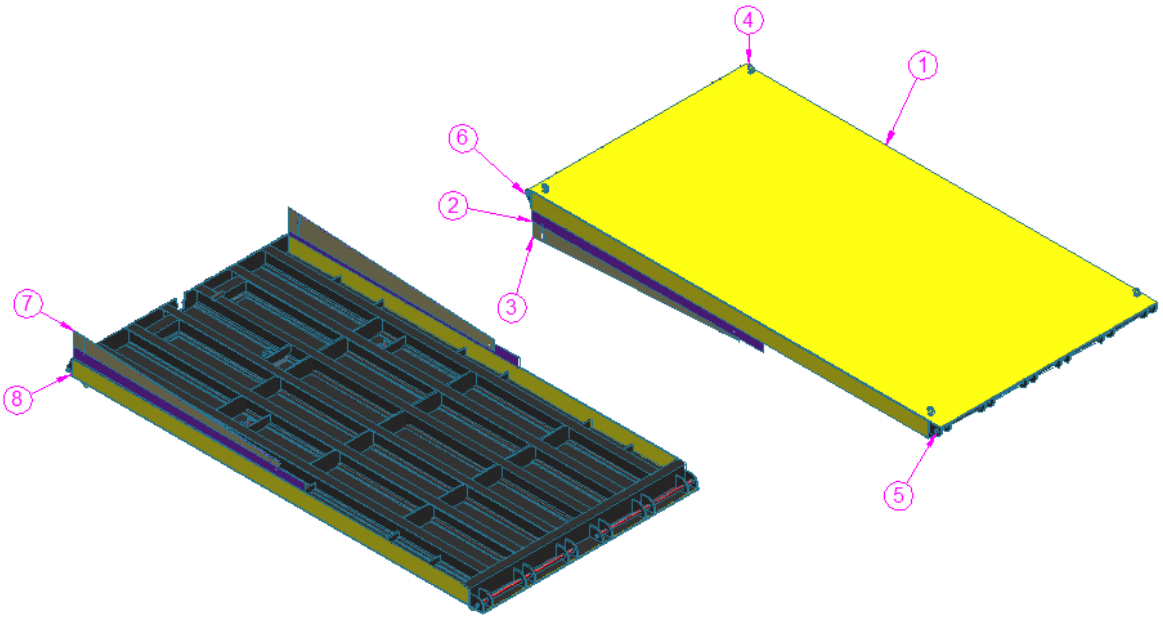
SR NO.	ITEM NO.	DESCRIPTION	QTY
1	170006010501	STOPPER LEVER	1
2	170006010502	LOCKING PIN	1

LIP UNIT



SR NO.	ITEM NO.	DESCRIPTION	QTY
1	170006010402	LIP-1	1
2	170006010404	LIP-2	1
3	170006010403	LIP-3	1
4	170006010401	LIP LOCK HANDLE	2

Top Frame Unit



SR NO.	ITEM CODE	DECRIPTION	QTY
1	170006010201	TOP FRAME	1
2	EYE BOLT M16	EYE BOLT M16	4
3	170006010204	Ø25 GUIDE BAR-1	2
4	170006010205	Ø25 GUIDE BAR -2	2
5	170006010203	SIDE COVER-2	2
6	170006010202	SIDE COVER-1	1
7	170006010206	SIDE COVER-3	1
8	170006010207	SQ TUBE 10X10	1

Safety Precautions - Dos and Don’ts

Do's of the system

- Power is switched off before entering into the Machine area for setting or maintenance.
- Press any E-Stop button (A red push button) and then follow the internal procedures of your company or organization for an emergency situation. Do not disconnect the power or Hydraulic supply to the system unless you are sure that nobody is trapped near the system. The E-Stop button is provided on Operator consol.
- Always ensure that system is” Switched off” while entering cell for maintenance or other activities.
- Use safety equipment’s like hand gloves etc.
- Use proper tools for the setting & maintenance activities.
- Clean the oil, dust & unwanted raw materials in the machine area after maintenance.

Don’ts of the system

- Do not set the system movements / programming with high speed.
- Do not bypass the limit switches of machines.
- Do not change the positions of the limit switches & dogs.
- Do not remove any covers.
- Do not lose the clamping screws on couplings, etc.
- Do not enter the machine when Auto cycle is ON.
- Do not alter the settings in the system.
- Do not put your hand while movement of torch slides while it is operating in given direction.
- Do not go away from the machine when machine is running.



– Note –

- Always maintain necessary logbooks for proper maintenance and Troubleshooting for trouble free production.

Preventive Maintenance

The importance of preventive maintenance cannot be over emphasized. Preventive maintenance will enable the machine to give long and trouble-free service.

Hydraulic Circuit of the machine

Sr. No.	Activity	Action	Daily	Weekly	Monthly	Half Yearly	Yearly
	Oil	Check		✓			
2	Oil Leakage	Check and Take Corrective Action	✓				
3	Hydraulic Pressure Switch	Check / Replace		✓			
4	Oil Filter	Clean / Replace Cartridge *		✓			

Hardware

SR. NO.	Activity	Action	Daily	Weekly	Monthly	Half Yearly	Yearly
1	Tightening of Nuts on Piston Rod	Check / Tighten			✓		
2	Sensors Sensitivity and Mounting	Check / Replace / Correct			✓		
3	a) Foundation Bolts / Leveling Bolts	Check / Tighten			✓		
	b) All Hardware Tightening	Check / Tighten				✓	
4	All parts in contact with the component (e.g. Resting blocks, Locating pins and Clamping pads.)	Check and replace		✓			

Lubrication

Sr. No.	Activity	Action	Daily	Weekly	Monthly	Half Yearly	Yearly
1	Oiling of all moving joints	Check			✓		

Preventive Maintenance for Lubrication

Hydraulic System Components

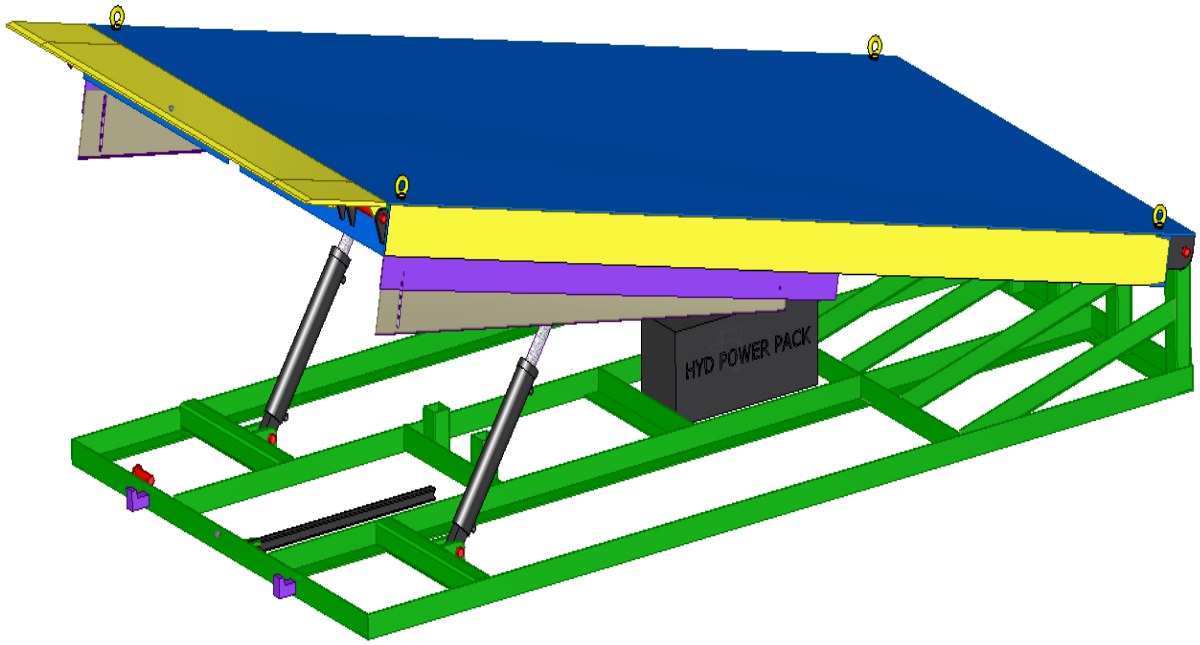
Subsystem	Observation	Cause	Solution
Hydraulic fittings.	Oil leakage.	Joints or cracked tubes. Loose fitting.	Check and replace tubes or other Hydraulic fittings. Proper fitting of all connectors.
Hydraulic cylinder.	Stroke of The cylinder insufficient.	Oil pressure low.	Adjust pressure setting.
		Oil leakage.	Check the Hydraulic circuit and Hydraulic fittings.
		Mounting arrangement not proper.	Do the correct mounting arrangement.
Solenoid operated valve	Cylinder not actuating.	Solenoid coil failure.	Replace the valve. Check the valve operation using manual override button on the valve.
		Electrical supply to valve is not correct.	Check the electrical supply.
		Hydraulic supply failure Solenoid failure due to sticky spool movement.	Check the hydraulic supply. Sometimes cleaning the valve assembly does help. However, this is a delicate and skilled job.
Flow Control Valve	Cylinder speed cannot be controlled satisfactorily	Flow control valve setting wrong.	Set the FCV correctly. Note that most of the Flow Control implementations are generally "Meter Out". In case of clamping application, the flow control implementations are "Meter In".
		Dust, dirt in the valve in the Hydraulic circuit.	Clean the FCV regularly.
	Oil leakage from FCV	Breakage of FCV	If defective then replace it.

Hydraulic faults & Troubleshooting. If a fault cannot be cured or if there is a serious mechanical or electrical failure, you should contact

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